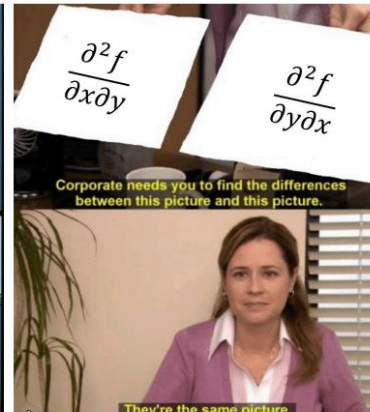
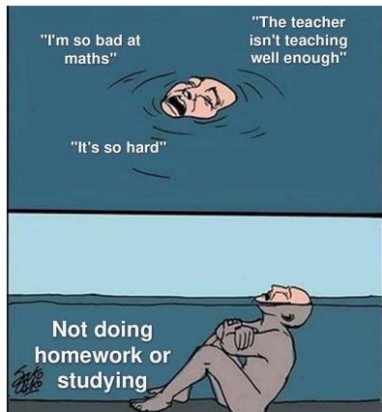


Agenda

- Recap
- Coding gradient descent
- Computing gradients of function
- Optimization problem
- Constrained optimization problem
 - How to convert constraint optimization problem into unconstrained optimization problem
 - Generalizing constraint optimization problem
 - Example of solving a constraint optimization problem
 - Solving the original constraint optimization problem
 - computing gradient of loss function with respect to w vector
 - computing gradient of loss function with respect to w_0
- Summary

Optimization 4: Constrained Optimization.

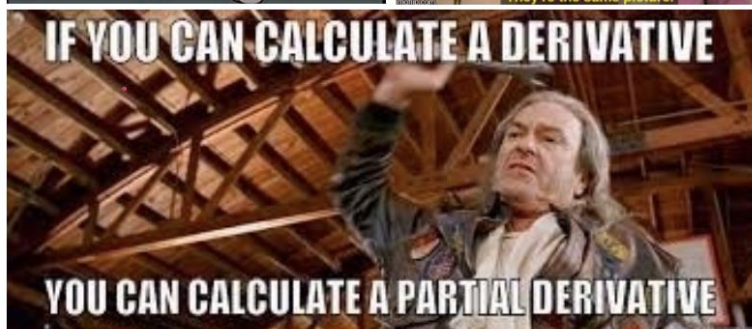


Recap:

- (a) Classification - choosing $\bar{w}$ and w_0 .
- (b) Brute force: very inefficient.
- (c) Alternative: gradient descent.
- (d) Functions, limits, derivatives.
- (e) Partial derivatives and gradients.

Today:

- (a) Applying Gradient descent to classification.
- (b) Lagrange Multipliers.
- (c) Constrained to unconstrained optimization.



Detailed Recap:

(a) Derivative: $\frac{d}{dx} f(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$

(b) Gradient descent update rule.

Problem: $\min_x f(x)$

$x^{(0)} \rightarrow$ Initial guess for x .

$$x^{(t+1)} = x^{(t)} - \eta \cdot \frac{\partial f(x)}{\partial x}.$$

(c) Partial derivative: $f(x, y, z).$

$$\frac{d}{dx} f(x, y, z) \quad ; \quad \frac{\partial}{\partial x} f(x, y, z).$$

$\rightarrow$ cannot treat y, z as constants.

$\nwarrow$ y, z are treated as constants.

Optimisation problem

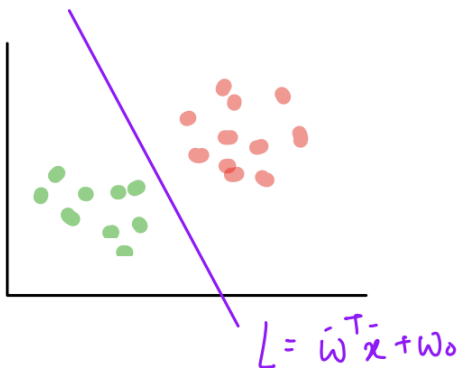
$$f(\underbrace{D}_{\text{dataset}}, \underbrace{\bar{w}, w_0}_{\text{parameters}}) =$$

$$\sum_{i=1}^n \left(\frac{\bar{w}^T \bar{x} + w_0}{\|\bar{w}\|} \right) y_i$$

$$D = \left\{ (\bar{x}_i, y_i) \right\}_{i=1}^n$$

$$\max_{\bar{w}, w_0} f(D, \bar{w}, w_0)$$

$$\min_{\bar{w}, w_0} \mathcal{L}(D, \bar{w}, w_0)$$



where

$$\mathcal{L}(D, \bar{w}, w_0) = -f(D, \bar{w}, w_0)$$

$$\min_{\bar{w}, w_0} \mathcal{L}(D, \bar{w}, w_0)$$

Unconstraint

problem we want to solve
using Gradient Descent.

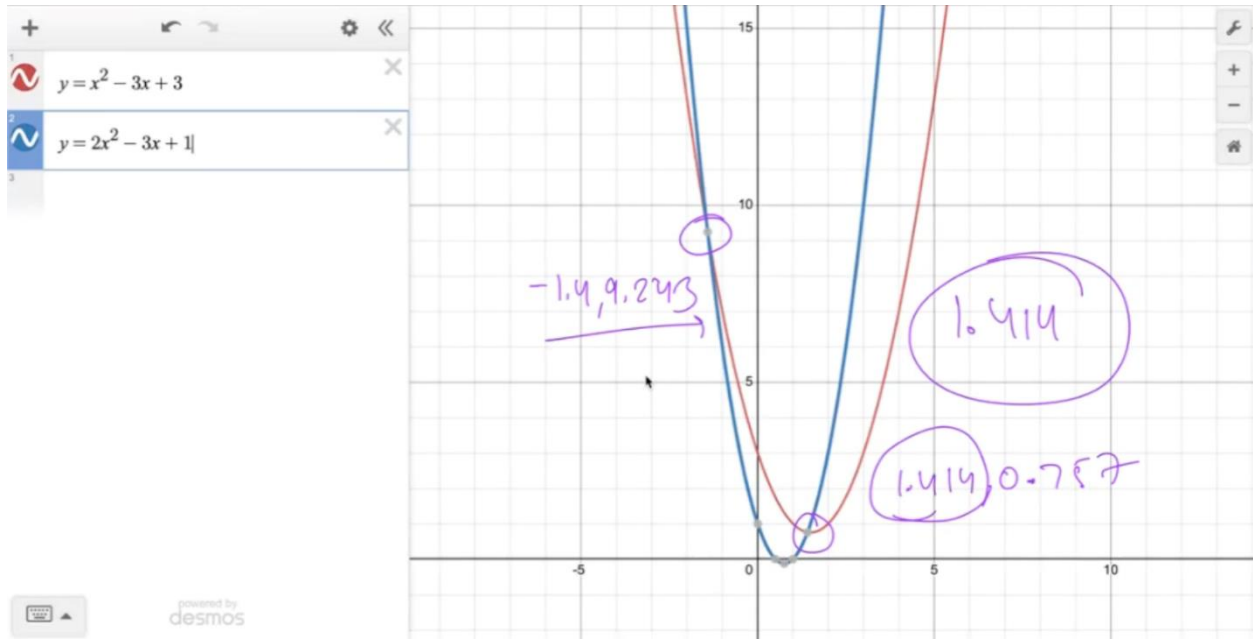
$$= \min_{\bar{w}, w_0} - \left(\sum_{i=1}^n \left(\frac{\bar{w}^T \bar{x} + w_0}{\|\bar{w}\|} \right) y_i \right)$$

$$= \min_{\bar{w}, w_0} - \left(\sum_{i=1}^n \left(\frac{\bar{w}^T \bar{x} + w_0}{\sqrt{w_1^2 + w_2^2 + w_3^2}} w_0 \right) y_i \right)$$

Convert: constraint optimisation problem.

It is difficult
to calculate Grad/
Deriv of this loss
function bcoz of
denominator.

we are only interested in
the direction of $\bar{w}$, we are
going to add a constraint to our
problem
 $\|\bar{w}\| = 1$



$$\min_{\bar{w}, w_0} - \sum_{i=1}^n (\bar{w}^T \bar{x} + w_0) y_i$$

$$\text{s.t. } \|\bar{w}\| = 1$$

(subject to)
(such that)

"Constraint optimisation problem"

How do I convert constrained optimisation problem into unconstrained optimisation problem?

"Lagrange multipliers".

Constrained
optimisation
problem



Lagrange
Multiplier



Unconstrained
optimisation
problem

$$\begin{aligned} \min_{\bar{x}} f(\bar{x}) \\ \text{st } g(\bar{x}) = 0 \end{aligned}$$

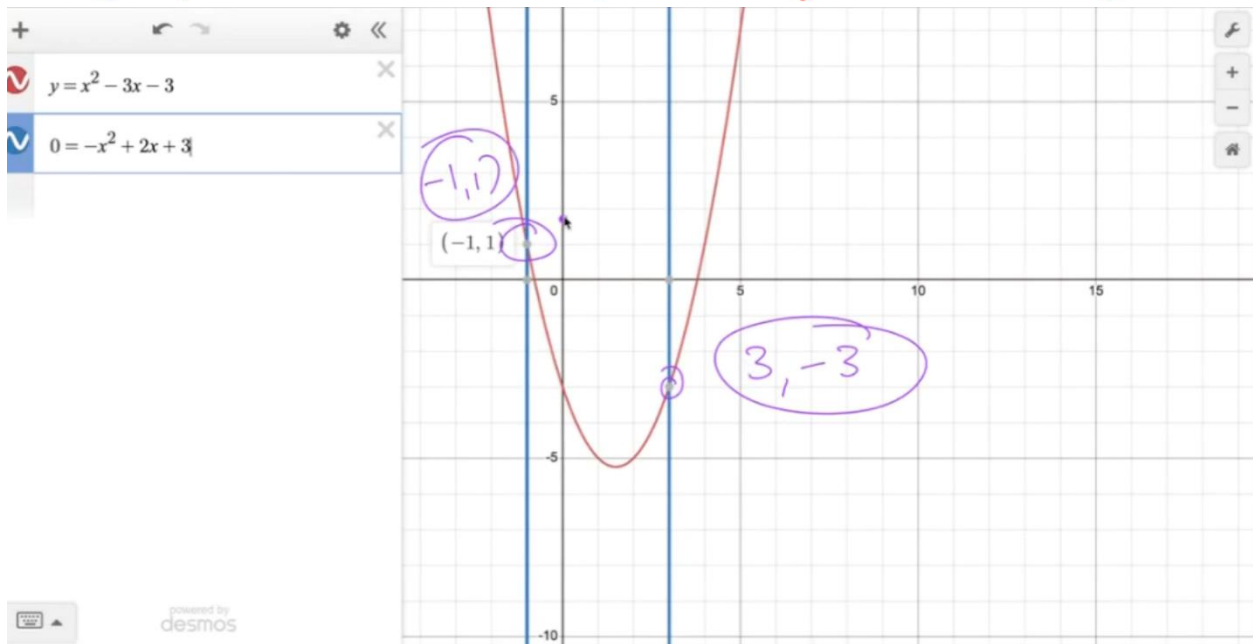
$$\min_{\bar{x}} f(\bar{x}) + \lambda(g(\bar{x}))$$

Generic $\min_{\vec{x}} f(\vec{x})$
 s.t. $g_1(x)=0, g_2(x)=0, g_3(x)=0, \dots, g_n(x)=0$

$$\min_{\vec{x}} f(\vec{x}) + \lambda_1 g_1(x) + \lambda_2 g_2(x) + \lambda_3 g_3(x) + \dots + \lambda_n g_n(x) = 0$$

Unconstrained optimisation problem (Lagrange Multiplier)

- ① First, we are going to count # Constraint Equation.
 - ② Introduce new variable for each constraints equation
 - ③ Rewrite the entire equation
- $$\min_{\vec{x}} f(\vec{x}) + \lambda_1 g_1(x) + \lambda_2 g_2(x) + \lambda_3 g_3(x) \dots \lambda_n g_n(x)$$



Q) Find minima of $f(x) = x^2 - 3x - 3$
 s.t. $-x^2 + 2x + 3 = 0$

Applying Lagrange multiplier

$$\left. \begin{array}{l} \min_{\bar{x}} f(\bar{x}) \\ \text{s.t. } g(x) = 0 \end{array} \right\} \rightarrow \min_{\bar{x}} f(\bar{x}) + \lambda g(x)$$

$$= \min_x x^2 - 3x - 3 + \lambda(-x^2 + 2x + 3)$$

$$= \min_x \boxed{x^2} - \boxed{3x} - \boxed{3} - \boxed{\lambda x^2} + \boxed{2x\lambda} + \boxed{3\lambda}$$

$$h(x, \lambda) \quad \nabla_{\bar{w}} h(\bar{w}) = \begin{bmatrix} \frac{\partial}{\partial x} h \\ \frac{\partial}{\partial \lambda} h \end{bmatrix} = \begin{bmatrix} 2x - 3 + 0 - 2\lambda x + 2\lambda + 0 \\ 0 + 0 + 0 - x^2 + 2x + 3 \end{bmatrix}$$

$$\begin{bmatrix} \textcircled{1} 2x - 3 - 2\lambda x + 2\lambda \\ \textcircled{2} -x^2 + 2x + 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Now, we have 2 variable with 2 equations.

$$2x-3 - 2\lambda x + 2\lambda = 0$$

$$2x-3 = 2\lambda x - 2\lambda$$

$$\frac{2x-3}{2(x-1)} = \lambda$$

$$\text{At } x=3, \lambda = \frac{2(3)-3}{2(3-1)} = \frac{3}{4}$$

$$\text{At } x=-1, \lambda = \frac{2(-1)-3}{2(-1-1)} = \frac{-5}{-4} = \frac{5}{4}$$

$$-x^2 + 2x + 3 = 0$$

$$-x^2 + 3x - x + 3 = 0$$

$$-x(x-3) - 1(x-3) = 0$$

$$(x-3)(x+1) = 0$$

$$x = 3, -1$$

By Solving the equation,
we now have candidate point for

$h(x, \lambda)$

$$(1) (3, 3/4)$$

$$(2) (-1, 5/4)$$

$$h(x, \lambda) = x^2 - 3x - 3 + \lambda(-x^2 + 2x + 3)$$

$$(1) = 3^2 - 3(3) - 3 + \frac{3}{4}(-3^2 + 2(3) + 3)$$

$$= \cancel{9} - \cancel{9} - 3 + \frac{3}{4}(\cancel{-9} + \cancel{6} + 3) = -3$$

Now, we are going to use Lagrange's Multiplier to make our problem easier. -

$$\min_{\bar{w}, w_0} - \sum_{i=1}^n (\bar{w}^T \bar{x} + w_0) y_i$$

$$\text{s.t. } \|\bar{w}\| = 1$$

$$\min_{\bar{w}, w_0} - \sum (\bar{w}^T \bar{x} + w_0) y_i + \lambda (\|\bar{w}\| - 1)$$

new problem that we have to solve.

$$\textcircled{2} \quad = (-1)^2 - 3(-1) - 3 + \frac{5}{4} \left(-(-1)^2 + 2(-1) + 3 \right)$$

$$= 1 + \cancel{3} - \cancel{3} + \frac{\cancel{5}}{4} \left(\cancel{-1} - \cancel{2} + \cancel{3} \right) = \textcircled{1}$$

$$h(x, \lambda) \quad (x, \lambda) \quad \begin{matrix} x & \lambda \\ (3, 3/4) \end{matrix} \rightarrow \begin{matrix} y \\ \textcircled{-3} \end{matrix}$$

$$(-1, 5/4) \rightarrow \textcircled{1}$$

Using Lagrange's Multiplier our minima point
(3, -3)

Now, we are going to use Lagrange's Multiplier to make our problem easier. -

$$\min_{\bar{w}, w_0} - \sum_{i=1}^n (\bar{w}^T \bar{x} + w_0) y_i$$

$$\text{s.t.} \quad \|\bar{w}\| = 1$$

$$\min_{\bar{w}, w_0} - \sum (\bar{w}^T \bar{x} + w_0) y_i + \lambda (\|\bar{w}\| - 1)$$

new problem that we have to solve.

$$\min_{\bar{w}, w_0} - \sum (\bar{w}^T \bar{x} + w_0) y_i + \lambda (\|\bar{w}\| - 1)$$

$$h(\bar{w}, w_0, \lambda)$$

$$(i) \quad \bar{w}^{(t+1)} = \bar{w}^{(t)} - \eta \underbrace{\nabla_{\bar{w}} h(\bar{w}, w_0, \lambda)}_{\text{Compute this}}$$

We are going to apply GD rule to update $\bar{w}$ & w_0

$$\min_{\bar{w}, w_0} - \sum (\bar{w}^T \bar{x} + w_0) y_i + \lambda (\|\bar{w}\| - 1)$$

$$h(\bar{w}, w_0, \lambda)$$

$$(i) \bar{w}^{(t+1)} = \bar{w}^{(t)} - \eta \underbrace{\nabla_{\bar{w}} h(\bar{w}, w_0, \lambda)}_{\text{Compute this}}$$

We are going to apply GD rule to update $\bar{w}$ & w_0

$$- \sum_{i=1}^n \left[\underbrace{(\bar{w}^T \bar{x}) y_i}_{\text{red}} + \underbrace{w_0 y_i}_{\text{green}} \right] + \lambda \left[\underbrace{\sqrt{\bar{w}^T \bar{w}}}_{\text{purple}} - \underbrace{1}_{\text{orange}} \right]$$

$$y_i \nabla_{\bar{w}} [\bar{w}^T \bar{x}] = y_i \bar{x}$$

$$\begin{aligned} & \nabla_{\bar{w}} [\lambda \sqrt{\bar{w}^T \bar{w}}] \\ &= \lambda \nabla_{\bar{w}} [\sqrt{\bar{w}^T \bar{w}}] \end{aligned}$$

$$= \lambda \times \frac{1}{2 \sqrt{\bar{w}^T \bar{w}}} \times \nabla_{\bar{w}} [\bar{w}^T \bar{w}]$$

$$= \lambda \times \frac{1}{2 \sqrt{\bar{w}^T \bar{w}}} \times \cancel{2} \bar{w}$$

$$= \lambda \times \frac{\bar{w}}{\sqrt{\bar{w}^T \bar{w}}} \Rightarrow \lambda \times \frac{\bar{w}}{\|\bar{w}\|}$$

$$f(x) = \sqrt{x}$$

$$h(x) = f(\bar{w}^T \bar{w})$$

$$f(h(x))$$

$$f(x) = x^{1/2}, f'(x) = \frac{1}{2} x^{1/2-1}$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{x}} = \frac{1}{2\sqrt{x}}$$

Putting everything together

$$\nabla_{\bar{w}} h(\bar{w}, w_0, \lambda) = -\sum_{i=1}^n y_i^2 \bar{x}_i + \lambda \frac{\bar{w}}{\|\bar{w}\|}$$

Summary

- * We wanted to build a classifier
- * We wanted to minimise a function

$$\alpha(D, \bar{w}, w_0) = -\sum_{i=1}^n \left(\frac{\bar{w}^T \bar{x}_i + w_0}{\|\bar{w}\|} \right) y_i^2$$

- * We converted this into

$$\min_{\bar{w}, w_0} (\bar{w}^T \bar{x} + w_0) y_i^2 + \lambda (\sqrt{\bar{w}^T \bar{w}} - 1)$$

using lagrange multiplier.

- * We got the GD update Rules

$$\bar{w}^{(t+1)} = \bar{w}^{(t)} - \eta \left[-\sum_{i=1}^n y_i^2 \bar{x}_i + \lambda \frac{\bar{w}}{\|\bar{w}\|} \right]$$

$$w_0^{(t+1)} = w_0^{(t)} - \eta \left[-\sum_{i=1}^n y_i^2 \right]$$